

FINAL DOSSIER

Inspection Order: IO-2025-1-M-38

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SECTION 5: PARTICULAR MATERIAL APPRAISAL (PMA)

PMA Records:

1. PMA Number: PMA-2025-001, Material Specification: ASME, Material Grade: SA-516 Gr 70, Status: active
Certified By: TUV NORD, Issue Date: 2023-06-20, Expiry Date: 2028-06-19

Werkstoff-Einzelgutachten (PMA)

gemäß Druckgeräterichtlinie (2014/68/EU), Anhang I, Nr. 4.2

Particular Material Appraisal (PMA) acc. to Pressure Equipment Directive (2014/68/EU), annex I, no. 4.2

SAP Nr. / SAP No.: 8115034257	PMA-Nr. / PMA-No.: 1326/W/1448/70/001		
Werkstoff / Material	Grade 70	Werkstoffgruppe/ Material Group/ISO 15608:	1.1
Druckgerätehersteller Manufacturer of the Pressure Equipment	THERMOPAC PROCESS ENGINEERING LLP. PLOT NO. B-20, MIDC TTC INDUSTRIAL AREA, CHINCHVALI, AIROLI, NAVIMUMBAI – 400708, INDIA		
Werkstoffspezifikation /Werkstoffnorm material specification/material standard	ASME SA 516 Section II Part A and Part D		
Prüfgrundlagen/Auslegungsnorm Requirements / Standard	PED 2014/68/EU, ASME Sec. VIII DIV. I		
Erzeugnisform / Abmessungsbereich Product form / Dimension range	Plate, Sheet / Thickness from 6 mm to 75 mm		
Anwendungszweck des Druckgerätes/Druckgeräteteils Application of pressure equipment / parts of pressure equipment	Flash Vessels, Gas Oil Evaporator, Receivers, Scrubber Tanks, HE Shells, Demister (Pressure Vessel / Equipment)		
Min./max. zul. Temperatur TS _{max} /TS _{min} Min./max. allowable temperature	- 29°C up to 340°C		

Zusätzliche Prüfungen/ Anforderungen / Additional tests / additional requirements :

- 1) Impact test from each heat treatment of a heat no. acc. to EN 148-1 at -29°C, requirement : KV ≥ 27J

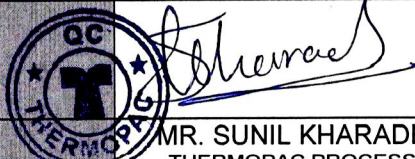
Bemerkungen / Remarks:

- 1) If the material manufacturer does not guarantee the material properties at service temperatures > 50°C, the pressure equipment manufacturer shall guarantee the qualification of the material for elevated temperatures (ASME BPVC II, Part D, Tab. Y-1 / Tab. 1A)
- 2) If the material manufacturer has not an appropriate quality assurance system (e.g. ISO 9001) with the material related assessments by a component body (see PED, annex I, sec 4.3), a test certificate 3.2 is necessary.

Ergebnis der Bewertung / Result of the evaluation:

The named material meets the Essential Safety Requirements (ESR) acc. to PED annex I, sec. 4 for the mentioned field of application.

Einschränkungen / Restriction: This approval valid until September 2022

	Druckgerätehersteller Pressure equipment manufacturer	Notifizierte Stelle 0045 Notified body 0045
Unterschrift / Signature		
Name / Name	MR. SUNIL KHARADE THERMOPAC PROCESS ENGINEERING LLP. NAVIMUMBAI – 400708, INDIA	Stefan Koch TÜV NORD Systems GmbH & Co. KG Große Bahnstraße 31 22525 Hamburg, Germany
Datum / Date		18.09.2017

Erfüllung der wesentlichen Sicherheitsanforderungen nach DGRL Anhang I, Nr. 4 Compliance with Essential Safety Requirements for materials, according to PED annex I, no. 4			
Eigenschaften / Property	Anforderungen / Requirement	Einzelheiten der Übereinstimmung Details of Compliance	erfüllt / fulfilled
Entsprechende Eigenschaften garantiert durch Werkstoffhersteller oder Druckgerätehersteller Appropriate properties assured by material manufacturer or Manufacturer of the Pressure Equipment (Anhang / Annex I - 4.1(a))	Dehngrenze/Streckgrenze (bei entsprechender Temperatur)/Zugfestigkeit Kriechdaten Proof strength/Yield strength (at appropriate temperature) UTS Creep data	at room temperature: acc. ASME Sec. II Part A SA 516 Gr. 70 at elevated temperature: acc. ASME Sec. II Part D SA 516 Gr. 70 see remarks: ¹⁾ creep is not expected	☒
Ausreichende Verformungseigenschaft Sufficiently ductile (Anhang / Annex I - 4.1(a)) (Annex I - 7.5)	Bei Stahl; A _{min} 14% for steel; A _{min} 14%	acc. ASME Sec. II Part A SA 516 Gr. 70 A ≥ 21%	☒
Ausreichende Zähigkeit Sufficiently toughness (Anhang / Annex I - 4.1(a), Annex I - 7.5)	z.B. bei Stahl; KV _{min} 27J bei 20 °C oder tiefster Betriebstemperatur. e.g. for steel; KV _{min} 27J at 20 °C or lowest operating temperature.	Impact Test: KV ≥ 27 J (- 29°C) additional tests / additional requirements: ¹⁾	☒
Keine wesentliche Beeinträchtigung durch Alterung Not significantly affected by ageing (Anhang / Annex I - 4.1(c))	Keine besondere quantitative Anforderung No specific quantitative requirement specified	aging is not expected due to manufacture	☒
Eignung für die vorgesehenen Verarbeitungsverfahren Suitability for intended processing (Anhang I / Annex I - 4.1(d))	Keine besondere quantitative Anforderung(en) No specific quantitative Requirement(s) specified	satisfactory forming and welding ability	☒
Werkstoffzeugnis Material Certification	Kat II, III & IV – Zeugnis über spezifische Prüfung des Produktes. Cat II, III & IV - Certificate of Specific Product Control.	DIN EN 10204: 3.1 Inspection Certificate see remarks: ²⁾	☒
Werkstoffkennzeichnung Material Marking	Rückverfolgbarkeit auf Schmelze/Los, Identifizierung des Werkstoffherstellers Traceability to heat/lot identity of the material manufacturer	acc. ASME Sec. II, Part A SA 516 Gr. 70	☒

Anlage/ Enclosure:



SECTION 6: TEST PROCEDURES

Test Procedure Records:

1. Procedure Number: TP-2025-001, Procedure Name: PENETRANT TESTING, NDT Method: PT, Applicable Standard:

Linked By: Current User, Linked Date: 7/27/2025

SECTION 4: MATERIAL TRACEABILITY

Material Records:

1. Certificate: 0000008367, Heat: 3416100, Grade: SA-516 Gr 70, Specification: ASME

INSPECTION CERTIFICATE																		BEW/Q & I/F/14	
CERTIFICATE NO.		BEW/ITC/242502414		TC Date :22/08/2024		EN 10204 3.1										Britex Engineering Works Plot No. R.209, T.T.C Industrial Area, M.I.D.C Rabale, Thane-Belapur Road, Navi Mumbai - 400701. 8657025614			
P.O. NO.		POR2425/462		PO Date :24/07/2024		Certificate No. : 0045/202/ 1409/WZ/0897/22 for PED. Certificate No. : 07/203/1409/WP/0897/22 for AD 2000 Merkblatt W0.													
CUSTOMER		Thermopac Process Eng.LLP																	
SOA :242500602		Plot No. B-20, MIDC, TTC Indl. Area, CHINCHVALI AIROLI Navi Mumbai																	
No	Description			PO.SR.NO.			Quantity			Heat Nos			Lot Nos						
1	FLANGE SA105 PN16 B1 DN50 [SORF] WITHOUT HUB			14			8			2302464			24H509						
No	Heat Nos	Chemical I.C. Ref. No.	Chemical Lab Name	%C	%Si	%Mn	%S	%P	%Cr	%Ni	%Mo	%Cu	%Cb	%V	%N	%Ti	%Al	%B	CEq
1	2302464	Heat Analysis(Raw Mat)	Rashtriya Isprit Nigam Limited	0.2000	0.1900	1.0900	0.0250	0.0220	0.0100	0.0010	0.0010	0.0010	---	0.0010	---	---	---	---	0.3840
2	2302464	Product Analysis RB9097	SOHM ANALYTICAL	0.2200	0.2200	1.2600	0.0250	0.0230	0.0100	0.0040	0.0080	0.0040	---	0.0010	---	---	---	---	0.4300
No	Heat Nos	Mechanical I.C. Ref. No.	Mechanical Lab Name	Tensile Test					Hardness HB				Charpy Impact Test V Notch (Joules) (-29°C Size:10x10x55mm				Grain Size	Heat Treatment	
				U.T.S N/mm2	Y.S. N/mm2 0.2 %	Y.S N/mm2 1 %	Elong- ation % G1=50mm	Reduction of area %	1	2	3	Avg	1	2	3	Avg	ASTM No.		
1	2302464	RB9097	SOHM ANALYTICAL	541.80	340.70	---	33.84	65.88	153	150	153	152	74	68	70	70.67	8	NORMALISED AT 920°C	
Note : - Punched on Outer dia: Size, Type, Pressure Rating, Specn. and Heat No., Lot No., BEW. on each flange. Remarks : 1. ABOVE MATERIAL CONFORMS TO ASME SA105 SEC.II PART-A ED-2021 & PED2014/68/EU. 2. Y.S = 0.2% OFFSET METHOD ELONGATION= ROUND SPECIMEN GAUGE LENGTH=L=4X DIA=4X12.5MM=50MM 3. DIMENSIONS OF ABOVE ITEM AS PER EN1092-1. 4. GASKET SEATING SURFACE FINISH FOUND WITHIN 125-250µN AARH.										Certified that the material described herein has been inspected and tested on requirements of the contract or purchase order.									
														 Ajay Singh Authorised Works Inspection					

Notes : 1) No part of this Test Certificate may be reproduced without the written permission of the QAM.



md. Anif. Khan.

SGS

WITNESS
REVIEWED

Date & Signature
SGS INDIA PVT LTD. THANE

The SGS stamp and signature are for use only when the stamp is placed on the SGS India Pvt Ltd. Stationery. The stamp is valid for use only when the stamp is placed on the SGS India Pvt Ltd. Stationery.

SECTION 7: SHOP INSPECTION RECORDS

Shop 1

Shop 2

SECTION 10: VISUAL INSPECTION RECORDS

VISUAL 1

SECTION 11: HYDROTEST REPORTS

Hydro 1

SECTION 14: ADDITIONAL SUPPORTING DOCUMENTS



SURYAADARSH CALIBRATION AND CONSULTANT (OPC) PVT. LTD.

Formerly UNIQUE CALIBRATION & CONSULTANT SERVICES

AN ISO 9001:2015 CERTIFIED CALIBRATION LABORATORY

NABL Accredited Calibration Laboratory As per ISO/IEC 17025:2017



CALIBRATION CERTIFICATE

Format No.: SCCPL-QF-7.8.1		Certificate No.: SCCPLCC20240911-002		ULR No.: CC328724000012733F	
Group: Pressure Indicating Devices			Discipline: Mechanical		
Customer's Name & Address Cali. Done Through : Bhagyalakshmi Analytical Solutions M/s. Thermopac Process Engineers LLP Navi Mumbai			Page No.: 1 of 1		Calibration Performed: At Lab
			Date of Receipt: 11-09-2024		
			Service Request No.: SCCPLSRN20240911-001		
			Certificate Issue Date: 13-09-2024		
			Date of Calibration: 12-09-2024		
Equipment Description (UUC) & Identification			Next Due Date (Suggested by Customer): 11-09-2025		
Name of the Instrument: Dial Pressure Gauge			Make: Airtech		
Range: 0 to 21 kg/cm ²			Model: -----		
Least Count: 0.5 kg/cm ²			Sr. No.: -----		
Condition of Instrument: Ok			I.D. No.: GAU-T-P-153		
Master Equipment Used			Location: -----		
Sr. No.	Instrument's Name	Range & I.D. No.	Calibration Agency	Calibration Certificate No.	Valid Up to
1	Digital Pressure Gauge	-1 To 70 bar & SCCPL/CAL/025	R & D Instrument Services	CS/24/LB/MP/118-01	16-05-2025
Environmental Conditions			Calibration Procedure	Calibration Reference Standard	
Temperature: 23 ± 1.5 °C, Humidity: 30% To 80% RH			SCCPL 03 SOP 007	DKD-R 6-1	

CALIBRATION RESULTS

Sr. No.	Set UUC Value (kg/cm ²)	Average Value of Ref. Standard (kg/cm ²)	Error (kg/cm ²)	Expanded Uncertainty (± % of rdg.)
1	0	0.000	0.000	-----
2	5	4.975	0.025	5.8
3	10	9.951	0.049	2.9
4	15	14.921	0.079	1.9
5	20	19.905	0.095	1.4
6	21	20.884	0.116	1.4

End of Certificate

Hemraj
Calibrated By

Hemraj
Calibration Engineer



Calibration done through:
Bhagyalakshmi Analytical
Solutions. Cell: 7900160492
Mail: blasgroup75@gmail.com

Reviewed/Approved By
L.M. Mahato

Chief Executive Officer (CEO)

- Note:** - 1. The calibration results reported in this certificate are valid at the time of and stated condition of measurement.
2. This report is not to be reproducing wholly or in part and cannot be used as evidence in the Court of Law and should not be used in any advertising media without our special permission in writing.
3. Total liability of our Organization is limited to the invoiced amount.
4. Calibration Certificate without signature are not valid.
5. All our certificates are metrological traceable.
6. The results relate only to the item calibrated.
7. UUC: Unit Under Calibration.
8. Expanded Uncertainty at 95 % confidence level with coverage factor K=2.
9. Scan this QR code to download our NABL Accreditation Scope & Certificate.



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